

NAME :FAHAD KHAN

ROLL NO: 23-SE-64

TASK 1:

```
import java.util.*;

public class ArrayTask1 {
    public static void main(String[] args) {
        int[] arr = new int[10];
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter 10 elements:");

        // Input
        for (int i = 0; i < 10; i++) {
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        int max = arr[0];
        int min = arr[0];

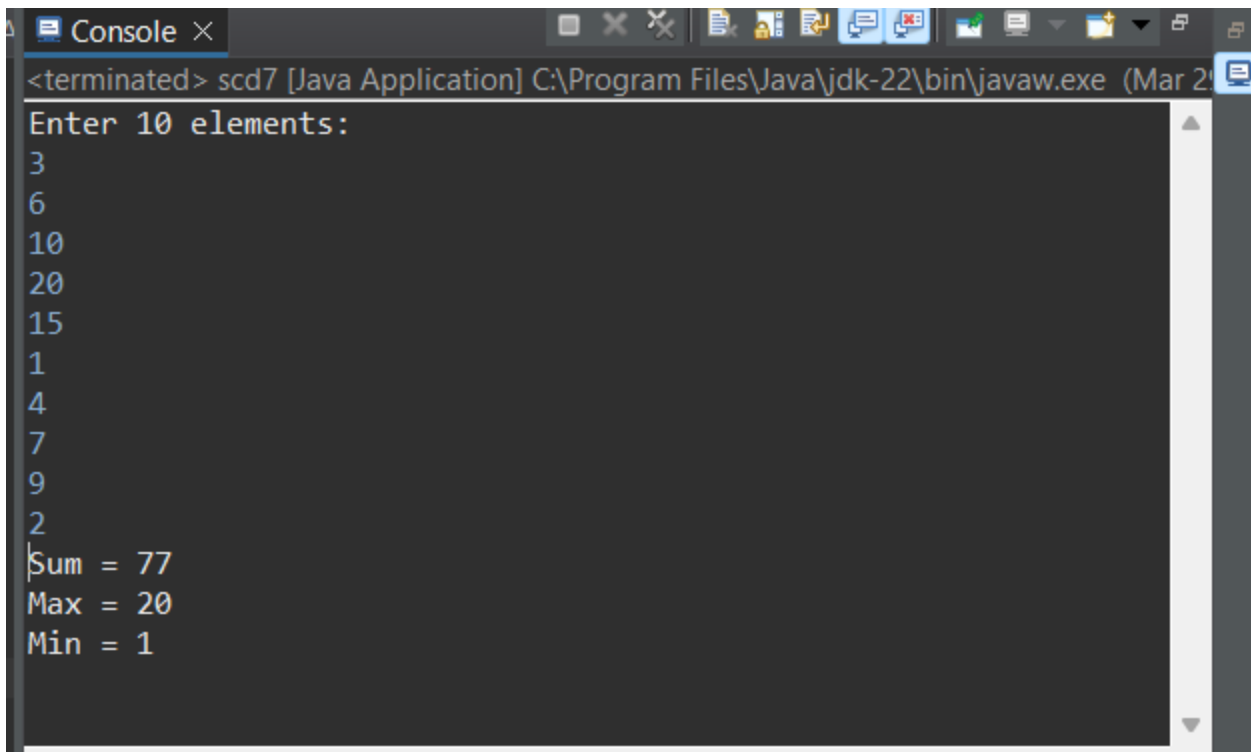
        // Processing
        for (int i = 0; i < 10; i++) {
            sum += arr[i];

            if (arr[i] > max)
                max = arr[i];

            if (arr[i] < min)
                min = arr[i];
        }

        // Output
        System.out.println("Sum = " + sum);
        System.out.println("Max = " + max);
        System.out.println("Min = " + min);
    }
}
```

OUTPUT:



```
<terminated> scd7 [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (Mar 2)
Enter 10 elements:
3
6
10
20
15
1
4
7
9
2
Sum = 77
Max = 20
Min = 1
```

TASK 2:

```
import java.util.*;

public class ArraySearch {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};
        Scanner sc = new Scanner(System.in);

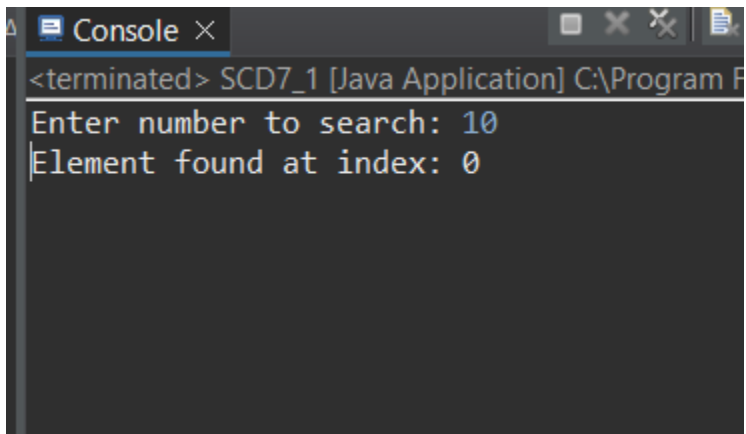
        System.out.print("Enter number to search: ");
        int key = sc.nextInt();

        boolean found = false;

        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == key) {
                System.out.println("Element found at index: " + i);
                found = true;
                break;
            }
        }

        if (!found) {
            System.out.println("Element not found");
        }
    }
}
```

OUTPUT:

A screenshot of a Java console window titled "Console". The window shows the output of a Java application. The first line is "<terminated> SCD7_1 [Java Application] C:\Program F". The second line is "Enter number to search: 10". The third line is "Element found at index: 0".

```
<terminated> SCD7_1 [Java Application] C:\Program F
Enter number to search: 10
Element found at index: 0
```

TASK 3:

```
class Node {
    int data;
    Node next;

    Node(int data) {
        this.data = data;
        this.next = null;
    }
}

public class LinkedListTask3 {
    Node head;

    // Insert at end
    public void insert(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    // Display list
    public void display() {
```

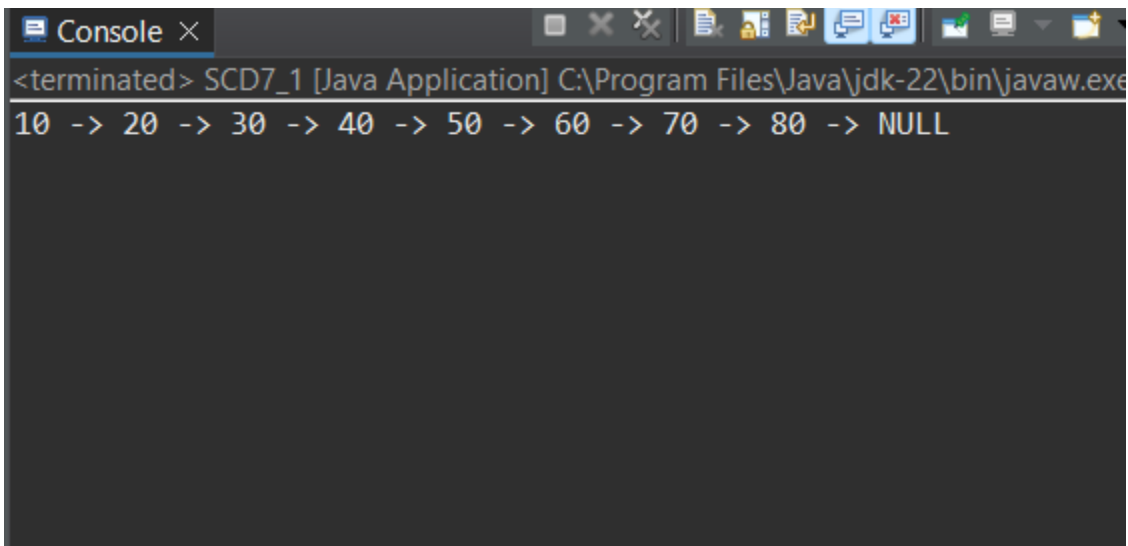
```
Node temp = head;
while (temp != null) {
    System.out.print(temp.data + " -> ");
    temp = temp.next;
}
System.out.println("NULL");
}

public static void main(String[] args) {
    LinkedListTask3 list = new LinkedListTask3();

    // Insert 8 nodes
    for (int i = 1; i <= 8; i++) {
        list.insert(i * 10);
    }

    list.display();
}
}
```

OUTPUT:



The screenshot shows a Java console window titled "Console" with a close button. The window displays the output of a Java application. The first line shows the command prompt: "<terminated> SCD7_1 [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe". The second line shows the output of the program: "10 -> 20 -> 30 -> 40 -> 50 -> 60 -> 70 -> 80 -> NULL".

```
<terminated> SCD7_1 [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe
10 -> 20 -> 30 -> 40 -> 50 -> 60 -> 70 -> 80 -> NULL
```

TASK 4:

```
public class LinkedListTask4 {
    Node head;

    // Insert at beginning
    public void insertBeginning(int data) {
        Node newNode = new Node(data);
        newNode.next = head;
        head = newNode;
    }

    // Insert at end
    public void insertEnd(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    // Delete a node
    public void delete(int key) {
        Node temp = head, prev = null;

        if (temp != null && temp.data == key) {
            head = temp.next;
            return;
        }

        while (temp != null && temp.data != key) {
            prev = temp;
            temp = temp.next;
        }

        if (temp == null) return;

        prev.next = temp.next;
    }

    // Display
```

```
public void display() {
    Node temp = head;
    while (temp != null) {
        System.out.print(temp.data + " -> ");
        temp = temp.next;
    }
    System.out.println("NULL");
}

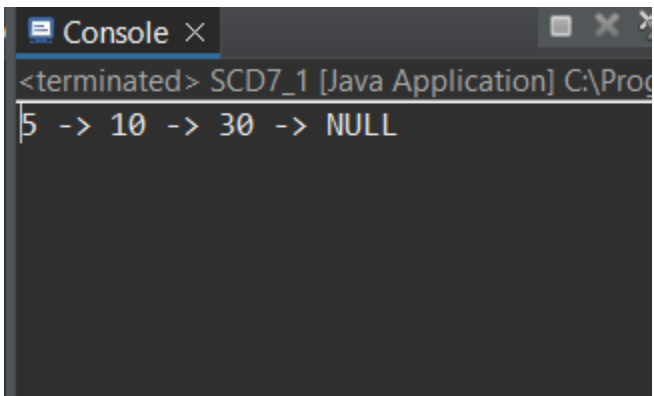
public static void main(String[] args) {
    LinkedListTask4 list = new LinkedListTask4();

    list.insertEnd(10);
    list.insertEnd(20);
    list.insertEnd(30);

    list.insertBeginning(5); // insert at start
    list.delete(20); // delete node

    list.display();
}
}
```

OUTPUT:

A screenshot of a Java IDE's console window. The window has a title bar with a tab labeled 'Console' and standard window controls. The main area of the console shows the output of the program: '<terminated> SCD7_1 [Java Application] C:\Pro...'. Below this, the output of the display method is shown as '5 -> 10 -> 30 -> NULL'.

```
<terminated> SCD7_1 [Java Application] C:\Pro...
5 -> 10 -> 30 -> NULL
```

TASK 5:

```
public class LinkedListTask5 {
    Node head;

    public void insert(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    // Count nodes
    public int count() {
        int count = 0;
        Node temp = head;

        while (temp != null) {
            count++;
            temp = temp.next;
        }
        return count;
    }

    // Find middle
    public void findMiddle() {
        Node slow = head;
        Node fast = head;

        while (fast != null && fast.next != null) {
            slow = slow.next;
            fast = fast.next.next;
        }

        System.out.println("Middle element: " + slow.data);
    }

    // Reverse list
    public void reverse() {
        Node prev = null;
```

```

Node current = head;
Node next;

while (current != null) {
    next = current.next;
    current.next = prev;
    prev = current;
    current = next;
}

head = prev;
}

// Display
public void display() {
    Node temp = head;
    while (temp != null) {
        System.out.print(temp.data + " -> ");
        temp = temp.next;
    }
    System.out.println("NULL");
}

public static void main(String[] args) {
    LinkedListTask5 list = new LinkedListTask5();

    list.insert(10);
    list.insert(20);
    list.insert(30);
    list.insert(40);
    list.insert(50);

    list.display();

    System.out.println("Total nodes: " + list.count());

    list.findMiddle();

    list.reverse();
    System.out.println("Reversed List:");
    list.display();
}
}

```

OUTPUT:


```
Console ×
<terminated> SCD7_1 [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (Ma
10 -> 20 -> 30 -> 40 -> 50 -> NULL
Total nodes: 5
Middle element: 30
Reversed List:
50 -> 40 -> 30 -> 20 -> 10 -> NULL
```